SMART PCB DEFECT DETECTION SYSTEM

B.Tech Mini Project Report

Development Platform: Base44 AI

Domain: Artificial Intelligence & Electronics

Branch: Electronics and Communication Engineering (ECE)

Project Link

https://app.base44.com/apps/6ac758a8b9298e7d69aec351/editor/preview

Student Name: _____

College: _____

Academic Year: 2026–2027

1. Abstract

Printed Circuit Boards (PCBs) are essential components of modern electronic systems. PCB quality directly affects the reliability and performance of electronic products. Traditional inspection can involve manual visual checking, which is time-consuming and may be affected by human error. The Smart PCB Defect Detection System is a web-based mini project developed using Base44 AI to demonstrate an AI-assisted approach to PCB image inspection. The application provides a user interface for submitting PCB images and presenting inspection information. The project demonstrates how AI-powered application development can be applied to an Electronics and Communication Engineering problem and provides a foundation for future integration with dedicated computer-vision and machine-learning models.

2. Introduction

The increasing complexity of electronic products has created a need for efficient PCB inspection. A PCB contains tracks, pads, components and other structures that must meet quality requirements. Defects can affect the operation and reliability of electronic devices. An AI-assisted inspection system can help organize and automate parts of the inspection workflow. This project focuses on creating a web application that demonstrates PCB image-based inspection using Base44 AI.

3. Problem Statement

Manual PCB inspection requires human effort and can be difficult to scale. Repetitive inspection may cause fatigue, and small visual abnormalities can be difficult to identify consistently. The project therefore proposes a web-based system that assists users in PCB image inspection and presents the inspection result in a clear format.

4. Objectives

• Develop a web-based PCB inspection application.
• Demonstrate the use of AI in an electronics-related application.
• Provide a simple interface for PCB image inspection.
• Assist in identifying possible PCB defects.
• Reduce repetitive manual inspection effort.
• Provide a foundation for future automated PCB quality inspection.

5. Proposed System

The proposed Smart PCB Defect Detection System accepts a PCB image through a web interface, performs the configured inspection/analysis workflow, and presents the result to the user. The current project is a prototype; production-level defect classification would require a validated computer-vision model and appropriate training/validation data.

6. System Workflow

USER → WEB APPLICATION → PCB IMAGE INPUT → IMAGE INSPECTION / AI ANALYSIS → DEFECT IDENTIFICATION → RESULT DISPLAY

7. Main Modules

User Interface: Provides the main interaction layer.
PCB Image Input: Allows a PCB image to be supplied for inspection.
Image Inspection: Processes the submitted image according to the application workflow.
Defect Analysis: Produces possible defect information based on the implemented logic/model.
Result Display: Presents the inspection outcome to the user.

8. Technology Stack

Base44 AI: AI-assisted application development platform.
Web Technologies: Application interface and interaction.
Artificial Intelligence: Intelligent application functionality.
Image Processing: PCB image inspection.
GitHub: Project documentation and version control.

9. Role of Base44 AI

Base44 AI was used as the primary application-development platform. The platform enables applications to be created and refined from natural-language requirements and supports application functionality, data capabilities and publishing. This makes it useful for rapid prototyping and student projects.

10. System Architecture

The system follows a simple user-to-result workflow:

User
↓
Web Application Interface
↓
PCB Image Upload/Input
↓
Image Processing / AI Analysis
↓
Defect Identification
↓
Inspection Result

11. Testing

Test Case 1: Provide a PCB image. Expected result: the application accepts the input and performs the configured inspection workflow.

Test Case 2: Provide a different PCB image. Expected result: the application processes the image and presents the corresponding inspection information.

Test Case 3: Provide invalid/unsupported input. Expected result: the application should show an appropriate validation or error message according to its implementation.

12. Advantages

• Simple web-based interface.
• AI-assisted development.
• Rapid prototyping.
• Reduces repetitive inspection effort.
• Can be extended with machine-learning models.
• Accessible through a web browser.
• Relevant to AI and ECE applications.

13. Limitations

• Detection performance depends on the implemented AI/image-analysis functionality.
• Image quality can affect results.
• Different PCB designs may require different datasets.
• A dedicated trained computer-vision model may be required for production-level classification.
• Industrial use requires extensive validation and testing.

14. Future Enhancements

1. Integrate a trained CNN/YOLO or suitable computer-vision model.
2. Classify multiple PCB defect categories.
3. Highlight defect regions using bounding boxes.
4. Display confidence scores.
5. Maintain inspection history.
6. Add analytics dashboards.
7. Generate downloadable inspection reports.
8. Integrate camera-based PCB inspection.
9. Connect with automated manufacturing and quality-control systems.

15. Applications

Potential applications include electronics manufacturing, PCB quality inspection, engineering laboratories, educational demonstrations, electronics research and smart-manufacturing prototypes.

16. Academic Relevance

The project combines Artificial Intelligence, Computer Vision, Image Processing, Electronics, PCB technology, Web Application Development and Software Engineering. It demonstrates the practical application of AI concepts to an ECE-related problem.

17. Recommended GitHub Structure

Smart-PCB-Defect-Detection-System/
■■■ README.md
■■■ documentation/
■ ■■■ mini-project-report.pdf
■■■ screenshots/
■ ■■■ home-page.png
■ ■■■ upload-page.png
■ ■■■ inspection-page.png
■ ■■■ result-page.png
■■■ sample-images/
■■■ LICENSE

18. Conclusion

The Smart PCB Defect Detection System demonstrates how an AI-powered web application can be applied to PCB inspection. The project provides a foundation for image-based inspection and can be enhanced with dedicated computer-vision models, defect localization, confidence scoring, inspection history and camera integration. Base44 AI enables rapid development and refinement of the prototype.

19. Project Link

Base44 Project Preview:
https://app.base44.com/apps/6ac758a8b9298e7d69aec351/editor/preview

20. References

1. Base44 AI App Builder – https://base44.com/ai-app-builder
2. Base44 Blog – How to Build an App with AI – https://base44.com/blog/how-to-build-an-app-with-ai
3. General concepts of PCB inspection, image processing and computer vision.